

2-layer FR-4 1.6 mm 1 oz Cu
Finish: ENIG (Ni 4 um / Au 0.075 um)
Mask: matte green / silk: white
Min clearance 0.15 mm / min trace 0.20 mm
Min via / drill 0.45 / 0.20 mm

Paste: TS391LT Sn42 Bi57.6 Ag0.4
 Stencil: 100 um SS (eC-stencil-mate)
 Bonder: Tresky T-3000-PR0
 Reflow: hot-plate, peak 165 °C
 Metrology: Keyence VK-X250 + X-ray + shear

Daniel Tyukov – student no. 5714699
ET4277 + ET4391 – TU Delft MSc EE
based on Abdelwahab et al. ECCE 2025

TIER-2 SOUTH PINOUT

1-4 D1 5-8 D2 9-12 D3 13-16 D4
17-20 D5 21-24 D6 25-28 D7 29-32 D8
each group: A, KG, KB, KR (A = LED VCC)
NORTH 3 10 19 32 = pre-wired chain (others inaccessible)